Revenue Retention Prep Guide

Staff Data Scientist, Core Revenue Retention - HighLevel

Built for a 1-2 day sprint. Assumes no hands-on survival analysis.

Chapter	Why it matters	Time
1. Role map	What they are really hiring for	10 min
2. Revenue retention metrics	GRR, NRR, mix effects. Day-1 fluency.	60 min
3. Churn types and recovery	Voluntary vs involuntary, dunning	30 min
4. Usage-based / CPaaS revenue	Churn when revenue decays, not cancels	30 min
5. Survival analysis from zero	Your biggest technical gap	90 min
6. Causal inference for retention	DiD, matching, synthetic control, traps	75 min
7. Add-on monetization	Attach rate, sizing, causality	30 min
8. Metric governance with Finance	One number that survives scrutiny	20 min
9. Interview cases	7 cases with model answers	90 min
10. Story bank and gap scripts	Your resume mapped to the JD	45 min
11. Self-check quiz	12 questions with answers	20 min

TWO-DAY PLAN
Day 1: Ch 1-5 (metrics, churn types, usage revenue, survival). Run the toy survival example by hand.
Day 2: Ch 6-8 morning, then Ch 9-11 afternoon. Say every model answer out loud once.
If you only have a few hours: Ch 2, Ch 5 sections 5.1-5.4, Ch 6 section 6.5, Ch 9 cases 1, 3, 4, then Ch 10.

EVIDENCE NOTE
Role facts come from the JD you shared. HighLevel-specific mechanics not in the JD are marked **[verify]**: treat them as hypotheses to ask about, not facts. Code samples were written but not executed here (the survival library was not installed), so run them once before the interview.

1. Role map

The JD asks you to **own one outcome: keeping and growing revenue from existing customers**, across teams that each hold a piece of it.

Surface	Who owns it	What you do
CPaaS (phone, SMS, email, WhatsApp)	Communications / CPaaS team	Largest MRR source. Diagnose churn and usage decay.
AI add-ons	AI product teams	Attach, usage, add-on revenue opportunity.
Customer Success	CS leaders	Which accounts to protect, whether outreach works.
Forecasts and definitions	Finance / RevOps	Canonical GRR/NRR, forecasting inputs.
Experiments	Experimentation lead	Test retention interventions you cannot randomize.

What 'Staff' means here

- **Direction-setting:** you decide how retention is measured company-wide, not only what the number is this week.
- **Standards others adopt:** reusable survival, DiD and synthetic control methods.
- **Influence without authority:** CS, Finance and CPaaS leaders act on your model.
- **Judgment on causality:** the JD wording is 'causal vs an artifact of how the data was generated'. That is the core test.

Four things the interviewer is checking

- Can you define GRR/NRR so Finance signs off?
- Can you tell real churn signal from mix, seasonality and data-maturity artifacts?
- Can you estimate causal effects when you cannot run an experiment?
- Has a diagnosis of yours changed a product, pricing, CS or lifecycle decision?

SAY IT IN THE INTERVIEW
"I read this role as owning one outcome, net and gross revenue retention, across CPaaS, add-ons, CS and Finance. My first move would be locking one certified definition with Finance, then separating real churn from mix and data artifacts before recommending anything."

2. Revenue retention metrics

2.1 The two core metrics

Pick a cohort: all customers paying at the **start** of a period (say, 1 Jan). Track only **that cohort's** MRR to the end of the period. New customers who joined mid-period are excluded.

Metric	Formula	Range	Reads as
GRR (gross revenue retention)	(Start MRR - churned MRR - contraction MRR) / Start MRR	0 to 100%	How much existing revenue you kept. Expansion is excluded.
NRR (net revenue retention)	(Start MRR - churned - contraction + expansion) / Start MRR	Can exceed 100%	Kept plus grown. Above 100% means the base grows with no new customers.

2.2 Worked example

A cohort starts the quarter at **$100,000 MRR** (100 customers).

Movement	MRR
Customers who cancelled (churn)	-$6,000
Customers who downgraded or used less (contraction)	-$3,000
Customers who upgraded or bought add-ons (expansion)	+$9,000

```
GRR = (100,000 - 6,000 - 3,000) / 100,000 = 91.0%
NRR = (100,000 - 6,000 - 3,000 + 9,000) / 100,000 = 100.0%
```

- NRR of 100% hides 9 points of leakage. **GRR shows the leak, NRR shows the net.** Always report both.
- Annualized vs quarterly: do not compare a quarterly 91% with an annual 91%. State the window every time.

2.3 Mix effects (the classic artifact)

NRR can move with **no behavior change in any segment**, only because the revenue mix changed. Example: NRR by segment is constant, but the share of Mid-market grows.

	SMB start MRR	Mid start MRR	SMB NRR	Mid NRR	Blended NRR
Q1	$40,000	$60,000	90%	110%	(36,000+66,000)/100,000 = **102%**
Q2	$20,000	$80,000	90%	110%	(18,000+88,000)/100,000 = **106%**

- Blended NRR rose 4 points. Neither segment changed. **Always decompose by segment before celebrating or panicking.**
- Fix: report segment-level NRR, or hold mix constant (standardize to a reference mix) when comparing periods.

2.4 Logo churn vs revenue churn

- **Logo churn:** % of customers lost. **Revenue churn:** % of MRR lost.
- They diverge when small accounts leave first: high logo churn, low revenue churn. The reverse signals a large account problem.
- For a business serving agencies and SMBs, segment by plan, size and agency vs direct before reading either.

2.5 Decomposition: how to answer 'NRR dropped'

Walk the bridge top-down. Each branch is a different team's problem.

```
NRR change
 |- GRR change
 |    |- Churn (cancellations)
 |    |    |- Voluntary (customer chose to leave)
 |    |    '- Involuntary (failed payment)         -> billing / dunning
 |    '- Contraction (downgrade, usage decline)   -> product / CS
 '- Expansion change
      |- Seat / plan upgrades                      -> pricing / sales
      '- Add-on attach and usage growth            -> AI + CPaaS teams
Then cut every branch by: segment, cohort (signup month), product, region, price change, channel.
```

TRAP / WATCH OUT
Cohorts matter. A drop in blended NRR may come from one bad signup cohort (for example, a heavy-discount acquisition month) rather than from the whole base. Cut by signup cohort before blaming product.

SAY IT IN THE INTERVIEW
"I'd report GRR and NRR together on the same fixed start-of-period cohort, always segmented, because blended NRR moves with mix even when no segment changes."

3. Churn types and recovery

3.1 Voluntary vs involuntary

	Voluntary	Involuntary
What it is	Customer decides to cancel or downgrade	Subscription ends because a payment failed
Typical causes	No value, price, competitor, business closed	Expired or replaced card, insufficient funds, bank decline, fraud rules
Recoverable?	Hard. Needs product or CS change.	Often yes, with the right billing workflow
Owner	Product, CS, pricing	Billing, RevOps, lifecycle messaging
How to detect	Cancel event or cancel survey	Payment-failed events followed by subscription end without a cancel action

- **First step is sizing:** what share of MRR churn is involuntary in your data? Do not assume. The answer sets priorities.
- Label quality is a data problem: if a failed payment leads to cancellation, some systems log it as a customer cancel. Check how billing events are recorded before trusting the split.

3.2 Dunning and recovery levers

Dunning means the process of recovering failed payments.

Lever	What it does	How to test it
Smart retry timing	Retry failed charges on a schedule instead of once (for example, different days, after common payday windows)	Randomize retry schedules across accounts, compare recovered MRR
Pre-dunning	Warn customers before a card expires	Holdout of accounts that get no warning
Card updater services	Card networks can refresh stored card numbers automatically	Compare failure rate for enabled vs not enabled (if staged)
Reminder emails and in-app banners	Prompt the customer to fix payment	Test copy, timing, channel
Grace period	Keep service on for N days after failure	Guardrail: revenue at risk vs recovery gain

HYPOTHESIS TO ASK ABOUT [verify]
HighLevel's usage products (phone, SMS, email) may be funded by a prepaid balance that auto-recharges. If so, a failed auto-recharge would be a distinct involuntary-churn trigger and usage would stop immediately. Ask: how is usage funded, and what happens on a failed recharge?

3.3 Metrics for recovery

- **Recovery rate** = recovered MRR / MRR that entered failed-payment state.
- **Time to recover** and **retry success by attempt number**.
- Guardrails: refunds, chargebacks, support contacts, customers annoyed by messaging.

SAY IT IN THE INTERVIEW
"Before building anything I'd size involuntary churn as a share of lost MRR, check how billing events are labelled, and then A/B the retry schedule and messaging with a holdout. It's the one churn type with a clean, testable fix."

4. Usage-based and CPaaS revenue

CPaaS (communications platform as a service) means customers pay per message, per call minute or per number. MRR here is **consumption**, so it moves even if no one cancels.

4.1 Why churn is harder to define

- A customer can keep the account open but send almost nothing. Revenue is gone, the logo remains: **silent churn**.
- Seasonality and campaigns create real usage spikes and dips unrelated to retention (an agency's client busy season).
- Revenue per unit changes with pricing, carrier cost pass-through and mix of channel (SMS vs voice vs WhatsApp).

4.2 Defining churn and contraction on usage data

Choose and document a rule. Example set of definitions (adapt with Finance and the CPaaS team):

State	Example rule	Why
Active	Revenue > $0 in trailing 30 days	Avoids counting a quiet week as churn
Hard churn	Account closed or no revenue for 60+ days	Clear, auditable
Soft churn / contraction	Trailing-90-day revenue down more than X% vs prior 90 days	Catches decay before cancellation. Choose X by looking at the historical distribution.
Reactivation	Revenue returns after hard churn	Counts separately from new, otherwise NRR is inflated

- Check how sensitive your conclusions are to the thresholds. If the story flips when 60 days becomes 45, say so.
- Smooth with trailing windows before classifying. Raw daily usage is noisy.

4.3 Drivers to decompose a usage revenue change

```
Revenue = Active accounts  x  Usage per active account  x  Price per unit
 - Active accounts down     -> logo churn / activation problem
 - Usage per account down   -> engagement or customer's own business slowing
 - Price per unit down      -> mix shift to cheaper channel, discounts, pricing change
```

TRAP / WATCH OUT
Compare like-for-like months. A holiday-heavy month has more messages for retail customers. Use year-over-year or seasonal adjustment before calling a trend.

SAY IT IN THE INTERVIEW
"For usage-based revenue I'd first agree with Finance on churn and contraction definitions, then decompose revenue into accounts, usage per account and price per unit, and test the thresholds for sensitivity."

HONESTY LINE
"I haven't worked in CPaaS. I have worked on engagement-driven churn and paid-subscription conversion, and I'd get up to speed quickly on the billing and usage data by walking the revenue bridge with Finance in week one."

5. Survival analysis from zero

5.1 The problem it solves

You want to know **how long customers last and what shortens it**. A simple 'churned within 12 months: yes/no' model has two problems:

- **Censoring:** customers who joined 4 months ago have not had 12 months to churn. They are not 'retained', their outcome is simply unknown yet.
- **Timing is thrown away:** churning in month 1 and month 11 look identical, though they mean very different things.

Survival analysis handles both. Vocabulary:

Term	Plain meaning
Duration	How long a customer was observed (months since signup)
Event	The thing you track: churn happened (1) or not (0)
Censored	Still active when the data ends, so duration is a lower bound, event = 0
Survival function S(t)	Probability a customer is still active at time t
Hazard h(t)	Risk of churning right now, given the customer survived until t
Hazard ratio (HR)	How a factor multiplies the hazard. HR 1.5 means 50% higher churn risk at any moment, holding other factors fixed.

5.2 Kaplan-Meier by hand (10-customer toy)

Ten customers. Duration in months, event 1 = churned, 0 = censored (still active or data ended).

Duration (mo)	2	3	3	5	6	6	8	9	12	12
Event	1	1	0	1	1	1	0	1	0	0

Rule: at each time where someone churns, multiply the running survival by (1 - churners / number at risk). Censored customers leave the at-risk pool without counting as churn.

Time	At risk	Churned	Censored	Step factor	S(t)
2	10	1	0	1 - 1/10 = 0.900	0.900
3	9	1	1	1 - 1/9 = 0.889	0.800
5	7	1	0	1 - 1/7 = 0.857	0.686
6	6	2	0	1 - 2/6 = 0.667	0.457
8	4	0	1	no churn, no change	0.457
9	3	1	0	1 - 1/3 = 0.667	0.305
12	2	0	2	no churn, no change	0.305

- **KM says 30.5% survive to month 12.** Naively counting 'not churned' gives 4 of 10 = 40%, because it treats the censored customers as retained. Naive overstates retention here.
- The curve steps down only at churn times. Censored customers shrink the denominator, which makes later steps bigger.

5.3 Kaplan-Meier in Python

```
# pip install lifelines  (not run in this guide's environment: run it once yourself)
import pandas as pd
from lifelines import KaplanMeierFitter

df = pd.DataFrame({
    "months":  [2,3,3,5,6,6,8,9,12,12],
    "churned": [1,1,0,1,1,1,0,1,0,0],
})
kmf = KaplanMeierFitter()
kmf.fit(df["months"], event_observed=df["churned"])
print(kmf.survival_function_)      # should match the hand table above
kmf.plot_survival_function()

# Compare segments
for name, g in df_all.groupby("plan"):
    KaplanMeierFitter().fit(g["months"], g["churned"], label=name).plot()
```

5.4 Cox proportional hazards (what drives churn)

KM compares groups. **Cox regression** estimates the effect of many factors at once on the hazard, without assuming a shape for the baseline curve.

```
from lifelines import CoxPHFitter

cph = CoxPHFitter(penalizer=0.01)           # small penalty helps stability
cph.fit(df_model[["months","churned","uses_phone","addon_ai","payment_failed",
                  "seats","tickets_90d"]],
        duration_col="months", event_col="churned")
cph.print_summary()                          # coef, exp(coef)=hazard ratio, p-value, CI
cph.check_assumptions(df_model, p_value_threshold=0.05)   # proportional hazards test
```

- **Reading it:** exp(coef) is the hazard ratio. payment_failed HR 2.3 means 2.3x churn risk at any time versus no failure, other things equal. addon_ai HR 0.7 means 30% lower hazard (association, not yet causal, see Ch 6).
- **Proportional hazards assumption:** an HR stays constant over time. If a factor matters a lot early and fades later, the assumption fails. Check it; fix by stratifying or by splitting time.
- **Time-varying covariates:** usage, tickets and payment failures change over a customer's life. Use a long-format table (one row per customer per period) and lifelines' CoxTimeVaryingFitter. This also reduces leakage versus using end-of-life values.
- **Evaluate:** concordance index (does the model rank who churns sooner?), calibration of predicted survival vs KM, and out-of-time validation.

5.5 When to use what

Question	Use
What does our retention curve look like? Compare plans or cohorts?	Kaplan-Meier + log-rank test
Which factors raise or lower churn risk over time?	Cox proportional hazards
Who will churn in the next 30 days (for CS)?	Cox or classifier on a rolling window. Survival gives time-to-churn, classification gives ranking.
Customers can churn then return?	Recurrent-event or competing-risk extensions. Mention, do not go deep.
Voluntary vs involuntary churn together?	Competing risks: model each cause-specific hazard separately.

TRAP / WATCH OUT
Survival models describe association. 'Add-on users have lower hazard' does not mean add-ons cut churn. Engaged customers adopt add-ons. Ch 6 covers how to get toward causal.

SAY IT IN THE INTERVIEW
"I'd use survival analysis because churn has censoring and timing: Kaplan-Meier for curves by segment, Cox with time-varying usage and billing features for drivers, and I'd check the proportional hazards assumption. Then I'd treat the hazard ratios as leads, not causes, until a causal design backs them."

6. Causal inference for retention

6.1 Why retention is full of causal traps

You rarely get clean experiments on pricing, CS outreach or add-on launches. Observational data is full of selection:

Trap	Example	Why it fools you
Self-selection	Customers who adopt an AI add-on retain better	Engaged, healthy customers adopt things. The add-on may do nothing.
Targeting bias	CS contacted at-risk accounts, who then churned more	CS picks risky accounts. Outreach looks harmful, or helpful if they pick wrong.
Survivorship	Long-tenure customers show high NRR	You only see the ones who stayed.
Reverse causality	Usage rises before a customer upgrades	Upgrade intent may cause the usage, not the other way.
Mix shift	NRR rose after a launch	Segment mix changed (Ch 2).
Seasonality	Churn dipped in Q4	Usage-driven customers are busier at year end.

6.2 First ask: can we randomize?

- **Yes:** run the experiment (dunning schedules, messaging, onboarding). the rest of this chapter covers what to do when you cannot.
- **Staged rollout:** a feature launched to groups at different times is a natural experiment (see DiD).
- **Sharp eligibility cutoff:** a threshold rule (for example, accounts above X usage get CS coverage) enables regression discontinuity.
- **No variation at all:** synthetic control or matching with sensitivity analysis, and be honest about limits.

6.3 Difference-in-differences (DiD)

Compare treated vs untreated groups, before vs after. The estimate removes anything that moved both groups the same way.

```
Effect = (Treated_after - Treated_before) - (Control_after - Control_before)

Example: monthly churn rate
              before    after
 Treated       4.0%      3.0%     change -1.0 pt
 Control       4.2%      3.9%     change -0.3 pt
 DiD effect = -1.0 - (-0.3) = -0.7 pt   (churn fell 0.7 pt more than it would have)
```

- **Key assumption: parallel trends.** Without treatment, the two groups would have moved together. Plot several pre-period months for both groups. Run an event-study (leads and lags) and expect pre-period coefficients near zero.
- **Staggered rollouts:** simple two-way fixed effects can be biased when effects differ across groups or over time. Know the names of better estimators (Callaway-Sant'Anna, Sun-Abraham) even if you do not implement them.
- **Spillover:** if control customers are affected by treatment (shared agency, shared community), the estimate is biased.

```
import statsmodels.formula.api as smf
# long format: one row per account-month; treated=1 for adopters; post=1 after launch
m = smf.ols("churned ~ treated*post + C(month)", data=df).fit(
        cov_type="cluster", cov_kwds={"groups": df["account_id"]})
print(m.params["treated:post"])        # DiD estimate; cluster errors by account
```

6.4 Matching and weighting

When treated and untreated differ on observable traits, match each treated account to similar untreated ones (or reweight with propensity scores), then compare outcomes.

- Match on **pre-treatment** variables only: tenure, plan, MRR, 90-day usage trend, past tickets, payment history. Never match on anything measured after treatment.
- Check balance after matching: standardized mean difference below about 0.1 on each variable.
- Check overlap: if treated accounts have no similar untreated twins, trim and say the result applies only to the overlap region.
- **The honest limit:** matching only controls for what you observe. An unmeasured 'intent to churn' variable still biases results. Run a sensitivity analysis (how strong would a hidden confounder need to be to erase the effect?).
- Combining matching with DiD (match on pre-trends, then difference) is stronger than either alone.

6.5 Synthetic control

Use when **one unit** is treated (one product line, one country, one customer tier) and no single clean comparison exists. Build a weighted average of untreated units that tracks the treated unit before the intervention.

- Weights are non-negative and sum to 1. Fit them on the **pre-period only**.
- After the intervention, the gap between actual and synthetic is the estimated effect.
- **Check pre-period fit.** If synthetic does not track the treated unit before launch, do not trust the post gap.
- **Placebo tests:** pretend each donor unit was treated and compute the same gap. If your effect is not unusual vs the placebo gaps, it is noise.
- Needs a long pre-period and a donor pool that resembles the treated unit. Weakness: too few donors or overfitting the pre-period.

6.6 Which method, when

Situation	Method	Main assumption to defend
Can randomize	A/B test with holdout	Clean assignment, no interference
Staged rollout, control group exists	DiD / event study	Parallel pre-trends
One treated unit, many donors	Synthetic control	Good pre-fit, placebo test passes
Eligibility threshold rule	Regression discontinuity	No manipulation around cutoff
Treated and untreated differ, rich pre-data	Matching / weighting (+ DiD)	No unobserved confounding (test with sensitivity)
Heterogeneous effects, rich data, no randomization	Double ML / causal forests (you have used DML)	Overlap, no hidden confounding

6.7 Worked example: did the AI add-on reduce churn?

- **Naive read:** add-on accounts churn 2 points less. This is selection until shown otherwise.
- **Step 1:** look for variation not driven by customer choice: a staged launch by region or plan, an eligibility rule, a random invite.
- **Step 2:** if staged rollout, DiD with pre-trend check. If not, match on pre-adoption usage, plan, tenure and MRR trend; then compare churn using survival with add-on as a time-varying flag.
- **Step 3:** look for dose-response (more usage, lower churn) and placebo outcomes (effect on things add-ons should not touch).
- **Step 4:** state the estimate with a range, name the assumptions, and say what experiment would settle it (randomized invite or holdout).

SAY IT IN THE INTERVIEW
"First I ask whether the variation is random. If it isn't, I look for a staged rollout or a threshold, check parallel trends or placebo tests, and I describe what a hidden confounder would need to look like to erase the result. If the decision is big, I recommend a holdout to close the gap."

7. Add-on monetization

7.1 Metrics

Metric	Definition	Use
Attach rate	Accounts with the add-on / eligible accounts	Penetration. Define 'eligible' carefully.
Time to attach	Days from signup (or from eligibility) to first purchase	Where in the journey the offer lands
Add-on usage per account	Units or revenue consumed per attached account	Real adoption vs buying and ignoring
Revenue per attached account	Add-on MRR / attached accounts	Value of the attach
Add-on churn	Share of attached accounts that drop the add-on	Product fit
Incremental NRR	NRR contribution of add-on expansion net of cannibalization	Does it add revenue or shift it?

7.2 Sizing the opportunity

A defensible sizing is eligible base x realistic attach x price, with a range and stated assumptions.

```
Example structure (numbers illustrative, not HighLevel data):
 Eligible accounts (use the add-on's real targeting rule)       = N
 Current attach rate                                            = a0
 Benchmark attach from best segment or comparable add-on        = a1   (range: a0 to a1)
 Price (or expected usage revenue per attached account)         = p
 Opportunity = N x (a1 - a0) x p x 12     (annualized, with low/base/high cases)
Then haircut for: cannibalization, discounting, adoption lag, onboarding capacity.
```

- Find the **drivers** of attach: which segments, which trigger events (hit a usage limit, onboarding step), which CS touchpoints. Prioritize segments with high eligible count and low attach.
- **Cannibalization check:** does the add-on replace base revenue (customer moves from per-message to a bundle)? Net the effect.

7.3 Do add-ons reduce churn?

- Same selection problem as Ch 6. Use staged rollout DiD, matched survival with add-on as a time-varying flag, and dose-response.
- Also check the reverse: low-churn customers have more budget and time to buy add-ons.

SAY IT IN THE INTERVIEW
"I'd size the opportunity as eligible base times realistic attach lift times price, show it as a range, haircut for cannibalization, and be explicit that 'add-on users retain better' is selection until a design shows otherwise."

8. Metric governance with Finance

The JD wants metrics that 'survive scrutiny from Finance and product alike' and canonical GRR/NRR on governed, certified data. Prepare a one-page definition checklist.

8.1 Definition checklist for GRR / NRR

Decision	Options to settle in writing
Source of truth	Billing system or ledger. Reconcile to Finance's recognized revenue monthly.
MRR definition	Subscription only, or plus recurring usage? Annual plans normalized to monthly?
Usage revenue	Trailing-30 or trailing-90 day average? How are spikes treated?
Cohort	Fixed start-of-period customers only. Exclude new logos from the numerator.
Credits, refunds, discounts	Net or gross? Timing of recognition.
FX	Reporting currency and rate date.
Reactivations	Counted as expansion, new, or separate line? Pick one.
Involuntary churn	When does a failed payment become churn? After how many days?
Window	Monthly, quarterly, annual. Never mix in one chart.
Segments	Plan, size band, agency vs direct, region, product (CPaaS, add-ons).

8.2 Operating model

- **You own the definition, Finance signs off, one certified dbt model serves everyone.** Dashboards and notebooks read from it, never recompute.
- Publish a metric glossary with the SQL, owner and edge-case rules. Version it; announce changes.
- Monthly reconciliation to Finance numbers, with a tolerance and a named explainer for any gap.
- Work with Analytics Engineering on the taxonomy (billing events, usage events, subscription state changes) rather than rebuilding pipelines. The JD says 'consume governed sources'.
- Standards for adjacent analysts: a short methods playbook (survival template, DiD template, synthetic control template, checklist of causal traps).

TRAP / WATCH OUT
Pipelines still under construction mean numbers can shift. Label data maturity in every read: what is certified, what is provisional, what is missing.

SAY IT IN THE INTERVIEW
"I'd lock a written GRR/NRR definition with Finance, publish it as a certified model, reconcile monthly to the ledger, and make every other team consume it. That ends metric debates in meetings."

9. Interview cases with model answers

Structure for every case: **clarify the decision, check data validity, decompose, form hypotheses, test, recommend, define how you'd know it worked.**

Case 1. NRR dropped 3 points this quarter. Walk me through it.

- **Is it real?** Check definition changes, billing or data pipeline changes, late-arriving data, one-off credits.
- **Same cohort basis?** Fixed start-of-period customers only, same window.
- **Bridge:** GRR vs expansion. Then churn vs contraction. Voluntary vs involuntary.
- **Segment and cohort:** plan, size, agency vs direct, signup cohort, product. Is the drop mix or behavior?
- **Seasonality:** compare year over year.
- **Hypotheses then tests:** pricing change, a payment-processing issue, a CPaaS pricing or deliverability change, a bad acquisition cohort, usage decay at large accounts.
- **Output:** a ranked list of drivers sized in MRR, owner for each, and a recovery plan.

SAY IT IN THE INTERVIEW
"I'd first confirm it is real and like-for-like, then walk the bridge, split every branch by segment and signup cohort, and only then form hypotheses. I'd end with the drop sized in MRR by driver and an owner for each."

Case 2. How would you build a churn model for CPaaS customers?

- **Define churn:** cancel vs usage decay (Ch 4). Agree on the label with Finance and CS.
- **Decision first:** who acts on it? CS contacts top-risk accounts weekly, so predict 30-60 days ahead.
- **Approach:** Cox or a discrete-time hazard model with time-varying features, or a classifier on rolling windows. Say why survival handles censoring.
- **Features:** usage trend, channel mix, payment failures, support tickets, add-on adoption, tenure, plan, deliverability or compliance issues.
- **Avoid leakage:** only data available at the prediction date. Out-of-time validation.
- **Evaluate:** concordance, precision at top-N accounts CS can handle, calibration, MRR captured in the top N (value-weighted).
- **Beyond prediction:** a high churn score does not mean outreach helps. Use a CS holdout to measure uplift (you have done this).

SAY IT IN THE INTERVIEW
"I'd define churn with Finance first, use a hazard model with time-varying usage and billing features, validate out-of-time, and rank by MRR at risk. Then I'd run a CS holdout, because who is likely to churn is not the same as who we can save."

Case 3. CS says accounts they contacted churned less. Did outreach work?

- **Selection:** how did CS pick accounts? If they chose engaged accounts, the comparison is biased in CS's favor. If they chose the worst, it is biased against.
- **Design options:** matching on pre-contact features and trends (and DiD on churn hazard), or use any quasi-random variation (CS capacity limits, alphabetical or territory assignment).
- **Going forward:** randomize a share of eligible at-risk accounts into a holdout. That is the clean answer.
- **Outcome:** retained MRR and net of CS cost, not only logo retention.
- **Heterogeneity:** which segments respond? Feeds a targeting rule.

SAY IT IN THE INTERVIEW
"Contacted accounts weren't randomly chosen, so the raw gap is selection. I'd match on pre-contact trends for a first read, and set up a holdout of eligible accounts so the next read is causal."

Case 4. Do AI add-ons reduce churn?

- Follow Ch 6.7: naive gap is selection; look for staged rollout or threshold; DiD with pre-trend check; matched survival with time-varying add-on; dose-response; placebo outcome.
- **Also consider the cost side:** AI add-ons have usage costs. Net contribution, not only retention.
- **Recommendation format:** estimate with range, assumptions, next experiment to settle.

SAY IT IN THE INTERVIEW
"The naive gap is selection until proven otherwise. I'd use any staged rollout for DiD, match on pre-adoption behavior otherwise, check dose-response, and recommend a randomized invite if the budget decision is large."

Case 5. How do you recover involuntary churn?

- Size it first: share of lost MRR that is involuntary, and check how billing labels it.
- Levers: retry schedule, pre-dunning for expiring cards, card updater, messaging, grace period (Ch 3).
- Test with randomization at account level. Metrics: recovery rate, time to recover. Guardrails: refunds, chargebacks, support contacts.
- Quantify the prize: recovered MRR x remaining lifetime.

SAY IT IN THE INTERVIEW
"It's the most testable churn type. I'd size it, check the labeling, randomize retry and messaging, and report recovered MRR with refund and chargeback guardrails."

Case 6. How do you get Finance and Product to agree on one number?

- Write the definition (Ch 8), reconcile to the ledger, and show the bridge from each team's number to the canonical one so no one feels overruled.
- One certified model, one owner, version history. Disagreements go to the doc, not the meeting.
- Start with the highest-stakes metric (GRR/NRR) and get Finance's sign-off before broadening.

SAY IT IN THE INTERVIEW
"I show each team the bridge from their number to the canonical one, get Finance's sign-off on the definition, and make the certified model the only source."

Case 7. A leader wants to cut a price or discount to reduce churn. What do you do?

- Clarify the target: which segment churns, and why? If churn is involuntary or product-fit, price will not help.
- Estimate elasticity from past price changes (DiD with a control segment), not naive regression.
- Net the trade: retained MRR minus lost price on the retained base. Survival-based projection of lifetime revenue.
- Pilot in a limited segment with a holdout before a broad change.
- Watch second-order effects: discount dependency, reference pricing, cannibalizing higher plans.

SAY IT IN THE INTERVIEW
"I'd first check price is actually the cause of churn in that segment, estimate elasticity from past changes with a control, net the lost price against retained revenue, and pilot with a holdout."

10. Story bank and gap scripts

10.1 Your resume mapped to the JD

JD ask	Your story (from your resume)	How to frame
Diagnosis that changed a decision	Supercoins earn-construct optimization and ring-fencing top customers: saved 10% of Supercoins cost. Netflix activation RCA: +50 bps month-end activations.	Lead with the decision and the number, then the analysis. State what you ruled out.
Churn and retention depth	Predictive churn models retaining 100K+ long-term purchasing customers monthly; at-risk test and control holdout.	Say what you built, how you proved incrementality, and what you'd add now (uplift layer, survival).
Causal inference judgment	Double ML for heterogeneous effects of Supercoins benefits where randomization wasn't feasible; Meta holdout design.	Explain why randomization failed, what you controlled for, what would break it.
Metrics Finance trusts	Single-source-of-truth dashboards; aligned Business, Finance and Product on 2025/2026 AOP.	Describe the definition fights you resolved and how you reconciled.
Subscription analogue	Black paid subscription conversion model; Netflix membership enrollment and activation; Plus Rehash measurement.	Position as paid-membership retention. Same shape as SaaS subscription retention.
Standards and mentoring	Loyalty Handbook, Activity Hub, Claude skills adopted by the pod; managing 3 analysts.	Show others use it without you.
AI-assisted analytics	Custom Claude skills cut RCA turnaround by 50%.	Be ready to say how you measured 50% and one thing it got wrong.

10.2 Gap-bridging scripts

SURVIVAL / HAZARD (gap)
"I've modelled churn with gradient-boosted classifiers and measured incrementality with holdouts. For this role I'd move to survival methods because customers join at different times and censoring matters. I've worked through Kaplan-Meier and Cox, including time-varying covariates and the proportional hazards check, and I'd pair them with a holdout for causal reads."
Only say 'worked through' if you actually run the Ch 5 code before the interview.

GRR / NRR / BILLING (gap)
"At Flipkart my revenue metrics were GMV and loyalty-program economics, not subscription MRR. The mechanics transfer: fixed-cohort retention, segment and mix decomposition, and reconciling to Finance. I'd spend week one with Finance walking the revenue bridge and the billing event definitions."

CPaaS / USAGE-BASED (gap)
"I haven't worked on telephony or messaging revenue. I have handled engagement-driven churn where there's no cancel event, so the first step is agreeing on a churn and contraction definition and testing sensitivity to thresholds."

SNOWFLAKE + dbt (minor)
"I've worked in BigQuery and Spark SQL with governed layers. The concepts transfer, and I'd pick up dbt conventions quickly."

10.3 Resume edits to consider

- Move the Black paid-subscription model and Netflix membership retention work higher; they read as 'subscription'.
- Add retained-revenue or retention numbers to the churn-model bullet if you have them. Do not invent them.
- Add 'survival analysis' only after you have actually used it.
- Fix the typo '1/22021' before the Senior Business Analyst line.

10.4 Questions to ask them

- How is CPaaS usage funded, and what happens on a failed payment? (billing mechanics)
- Which data sources are certified today for MRR, and what is the plan for the rest?
- How is churn defined today for usage-based accounts, and who owns that definition?
- What past retention decision did the data team change, and what stopped it from being faster?
- How does the Experimentation lead split work with this role?

11. Self-check quiz

Cover the answers. Aim for crisp one-liners.

Q1. A cohort starts at $200k MRR. Churn $10k, contraction $6k, expansion $20k. GRR and NRR?

A: GRR = (200-10-6)/200 = 92%. NRR = (200-10-6+20)/200 = 102%.

Q2. Why can blended NRR rise with no segment improving?

A: Mix shift toward segments with higher NRR (Ch 2.3).

Q3. Why exclude new customers from NRR?

A: NRR tracks only the fixed start-of-period cohort. New logos would inflate it.

Q4. What is censoring and why does it matter?

A: A customer still active at data end has unknown churn time (event = 0). Treating them as retained or churned biases retention estimates.

Q5. Hand check: why is KM at month 12 (30.5%) lower than the naive 40%?

A: Naive counts censored customers as retained. KM removes them from the at-risk pool at their censoring time, so later churn steps are larger.

Q6. Hazard ratio 0.7 for an add-on flag: meaning and caveat?

A: 30% lower churn hazard at any time, other covariates fixed. Association only: customers who adopt differ from those who don't.

Q7. What does the proportional hazards assumption say, and how do you check it?

A: A covariate's effect (HR) is constant over time. Check Schoenfeld-residual tests (lifelines check_assumptions) or plots; fix with stratification or time splits.

Q8. Parallel trends: how do you defend it?

A: Plot multiple pre-period months for both groups, run an event study with leads near zero, and use matched controls.

Q9. Synthetic control: what two checks before trusting it?

A: Good pre-period fit, and placebo tests applying the method to donor units.

Q10. CS says contacted accounts retained better. First objection?

A: Selection: CS chose which accounts to contact. Fix with matching/DiD now and a randomized holdout going forward.

Q11. How would you size an add-on opportunity?

A: Eligible base x realistic attach lift x price, annualized, as a range, with haircuts for cannibalization and adoption lag.

Q12. What must be written down to make GRR/NRR survive Finance scrutiny?

A: Source of truth, MRR and usage treatment, fixed cohort, credits and refunds, FX, reactivation treatment, involuntary churn rule, window, segments.

FINAL 10-MINUTE CHECKLIST
1) Say the GRR/NRR formulas and the mix example from memory. 2) Redo the KM table on paper. 3) State the three selection traps with examples. 4) Recite your three gap scripts. 5) Have one real story for each of: decision changed, churn model with holdout, causal read without randomization.